

	Second order maxima would be observed at $(180 \pm 36)^\circ = 238^\circ$
15.	Ans: B Rayleigh's criterion: minimum angle for image to be just resolved, $\theta = \lambda/b$ $\sin \theta \approx \tan \theta$ $\Rightarrow \lambda/b = x/9.0 \Rightarrow 500 \times 10^{-9}/25 \times 10^{-3} = x/9.0$ $x = 0.18 \text{ mm}$
16.	Ans: B In a closed loop, change in internal energy, $= 0$